

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Date: Aug 24, 2023

Pre-Tutorial Assignment 3

Marks: 2

1. Figure 1 shows an electric circuit with independent voltage and current sources. The numerical values of the circuit elements are as follows: $V_1 = 10V$, $V_2 = 5V$, $I_1 = 1A$, $R_1 = 1\Omega$, $R_2 = 2\Omega$, and $R_3 = 3\Omega$. Using mesh analysis, evaluate the potential difference V_{I1} across the current source I_1 .

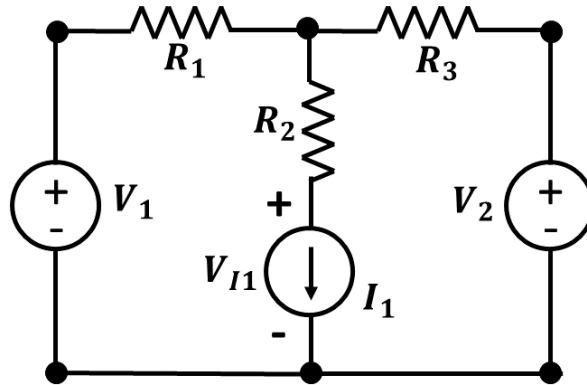


Figure 1: Electric circuit with independent voltage and current sources.

2. Consider a four bit number $(ABCD)_2$ which has a value between 0 and 15, both inclusive. A digital circuit needs to be built which would recognize if the input is a multiple of $(11)_2$. Define the function $F(A, B, C, D)$ so that $F = 1$ if and only if the number $(ABCD)_2$ is divisible by $(11)_2$.
- Express the function $F(A, B, C, D)$ in the Standard SOP form.
 - Express the function $F(A, B, C, D)$ in the Standard POS form.